

2021 Second Round of Nepal Mathematics Olympiad

October 30, 2022

Problem 1

In triangle ABC let X and Y lie on the side AB and AC such that $BXCY$ is concyclic and XY intersect the circumcircle of triangle ABC at P . If $AX : BX = 1 : 8$ find $XP : BP$.

Problem 2

Show that

$$\left(a + 2b + \frac{2}{a+1}\right) \left(b + 2a + \frac{2}{b+1}\right) \geq 16$$

for all positive numbers a and b such that $ab \geq 1$

JBMO 2013 Problem 3

Problem 3

Find all triples (p, q, r) of prime numbers such that all of the following numbers are integers $\frac{p^2+2q}{q+r}, \frac{q^2+9r}{r+p}, \frac{r^2+3p}{p+q}$

JBMO Shortlist 2019

Problem 4

In a certain city there are n straight streets, such that every two streets intersect, and no three streets pass through the same intersection. The City Council wants to organize the city by designating the main and the side street on every intersection. Prove that this can be done in such way that if one goes along one of the streets, from its beginning to its end, the intersections where this street is the main street, and the ones where it is not, will appear in alternating order.
JBMO Shortlist 2019